

Topic: Join Operation

Q. Explain the Join operation in Relational Algebra. Describe the different types of Joins including Theta Join, Equi Join, Natural Join, and the Outer Joins.

> **Definition to Remember**

Join Operations ()

(Return only rows with a matching counterpart in both tables)

A. Theta Join ({}_{})

B. Equi Join

C. Natural Join ()

2. Types of Outer Joins

(Preserve unmatched rows missing values are filled with NULL)

...

Roll | Name Roll | Course

| Join Type | What is Returned |

Example (Left Outer Join): STUDENT Left Join ENROLLMENT Shows ALL students. Luke has no enrollment his Course column = NULL.

3. Summary: All Join Types

| Join Type | Returns | Duplicates |

| :--- | :

> **Must-Write Points (for 10 marks)**

> 1. Jo

> **Quick Recall**

> `Join
Left/Right